

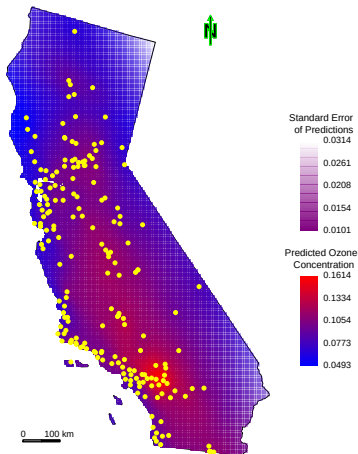
An Introduction to Statistical Models for Spatial Data in Ecology

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National Marine Mammal Lab
NOAA Fisheries Alaska Science Center
International Arctic Research Center
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Outline

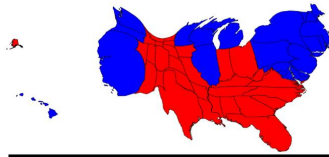
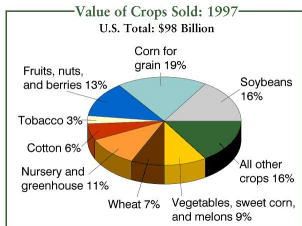
- ▶ Introduction
- ▶ Types of Spatial Data
- ▶ Types of Inference
- ▶ Model Selection
- ▶ Summary

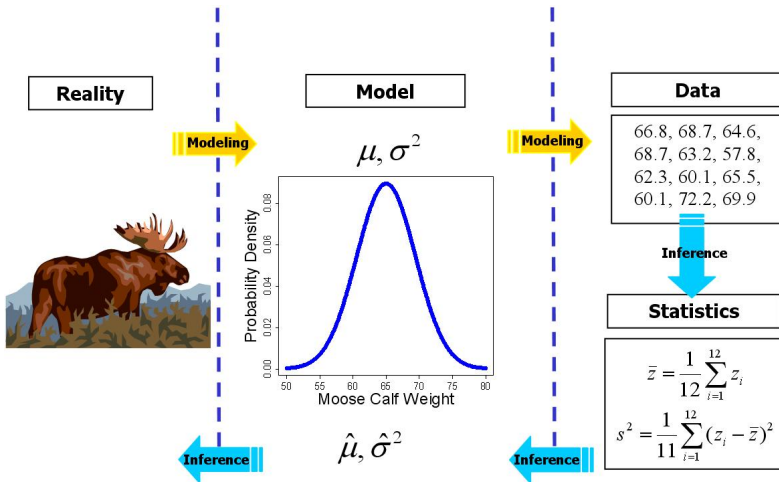


What are Statistics?

$$\bar{z} = \frac{1}{n} \sum_{i=1}^n z_i$$

A statistic is a function of data





$$z_i = \beta_0 + x_{1,i}\beta_1 + x_{2,i}\beta_2 + \dots + \epsilon_i$$

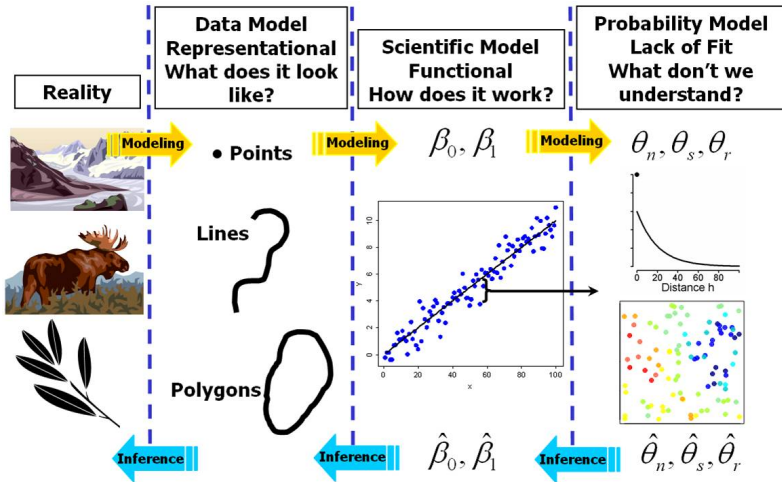
$$z_{j,k,i} = \beta_0 + \tau_j + \delta_k + (\tau\delta)_{j,k} + \dots + \epsilon_{j,k,i}$$

$$\begin{pmatrix} \mathbf{z}_{\text{observed}} \\ \mathbf{z}_{\text{unobserved}} \end{pmatrix} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \text{ var}(\boldsymbol{\epsilon}) = \boldsymbol{\Sigma}(\boldsymbol{\theta})$$

- ▶ Point Prediction
- ▶ Block Prediction
- ▶ Sampling

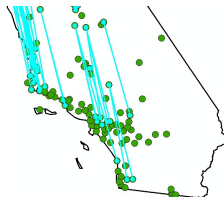
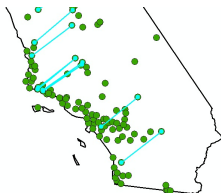
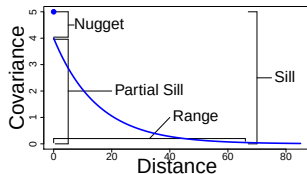
- ▶ Regression
- ▶ Design of Experiments

Statistical Models and Inference



Reduce the number of parameters in the probability model by using spatial relationships: AUTOCORRELATION

$$\Sigma = \begin{pmatrix} \sigma_{1,1} & \sigma_{1,2} & \cdots & \sigma_{1,n} \\ \sigma_{2,1} & \sigma_{2,2} & \cdots & \sigma_{2,n} \\ \vdots & \vdots & \cdots & \vdots \\ \sigma_{n,1} & \sigma_{n,2} & \cdots & \sigma_{n,n} \end{pmatrix}$$



Spatial Generalized Linear Mixed Models

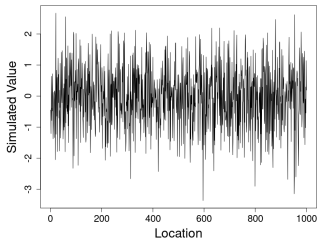
$$\begin{pmatrix} \mathbf{z}_{\text{observed}} \\ \mathbf{z}_{\text{unobserved}} \end{pmatrix} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}_1\mathbf{u}_1\delta_1 + \dots + \mathbf{Z}_q\mathbf{u}_q\delta_q + \boldsymbol{\epsilon}, \text{ var}(\boldsymbol{\epsilon}) = \boldsymbol{\Sigma}(\boldsymbol{\theta})$$

$$f\left(\begin{pmatrix} \mathbf{z}_{\text{observed}} \\ \mathbf{z}_{\text{unobserved}} \end{pmatrix}\right), \quad g[E(\mathbf{z})] = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}_1\mathbf{u}_1\delta_1 + \dots + \mathbf{Z}_q\mathbf{u}_q\delta_q + \boldsymbol{\epsilon}$$

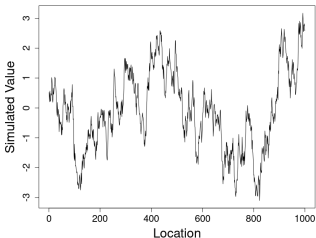
- ▶ Point Prediction
- ▶ Block Prediction
- ▶ Sampling
- ▶ Regression
- ▶ Design of Experiments

Autocorrelated Errors

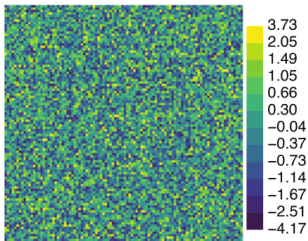
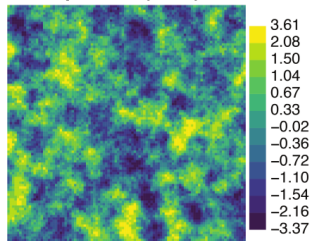
Independent Errors



Autocorrelated Errors

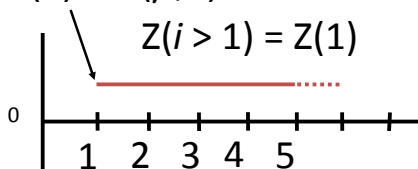


No-correlation

Spherical ($\alpha = 1$)

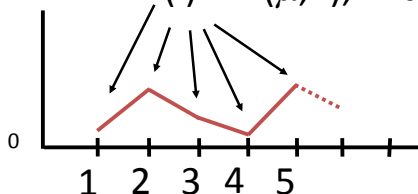
Estimation Versus Prediction

$$Z(1) \sim N(\mu, 1)$$

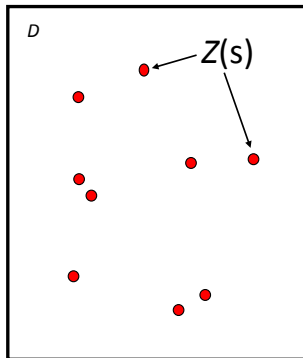


$$\Sigma = \begin{pmatrix} 1 & 1 & \dots & 1 \\ 1 & 1 & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \dots & 1 \end{pmatrix}$$

$$Z(i) \sim N(\mu, 1), \text{ i.i.d.}$$



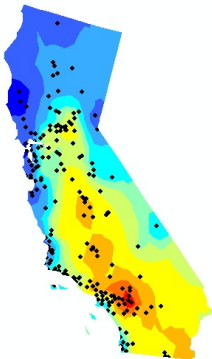
$$\Sigma = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$



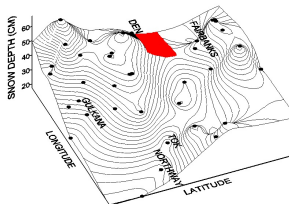
- ▶ D is the spatial domain or area of interest
- ▶ s contains the spatial coordinates
- ▶ Z is the value located at the spatial coordinates

Examples of Geostatistical Data

Ozone Predictions

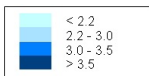
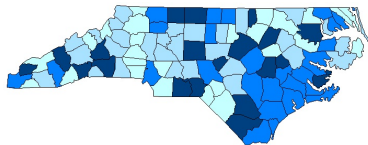


Snow Depth Predictions

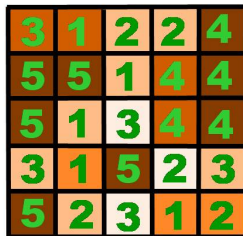


Examples of Lattice/Aerial Data

Transformed SIDS rates

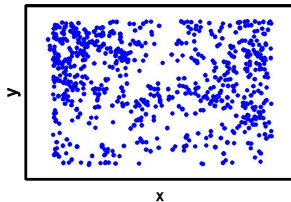


Plots in a Designed Experiment

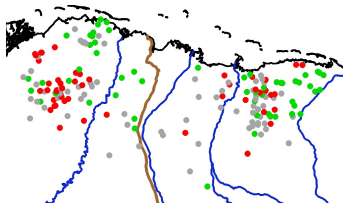


Examples of Point Pattern Data

Lansing Woods
Hickories Locations



Arctic Caribou
Calving Locations



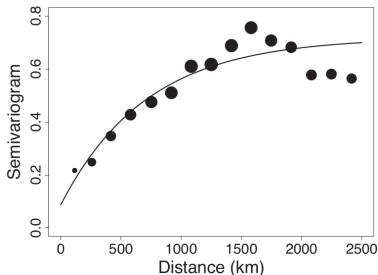
Covariance Parameter Estimation

$$\begin{pmatrix} \mathbf{z}_{\text{observed}} \\ \mathbf{z}_{\text{unobserved}} \end{pmatrix} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \text{ var}(\boldsymbol{\epsilon}) = \boldsymbol{\Sigma}(\boldsymbol{\theta})$$

- fitting variograms and covariance functions

$$\hat{\gamma}(r_k) = \frac{1}{2N_k} \sum_{\|\mathbf{s}_i - \mathbf{s}_j\| \in R_k} (y_i - y_j)^2 \quad (k = 1, \dots, K)$$

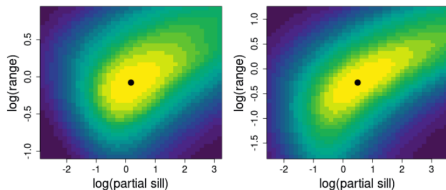
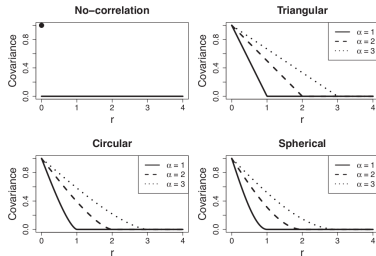
$$\hat{C}(r_k) = \frac{1}{N_k} \sum_{\|\mathbf{s}_i - \mathbf{s}_j\| \in R_k} (y_i - \bar{y})(y_j - \bar{y}) \quad (k = 1, \dots, K)$$



Covariance Parameter Estimation

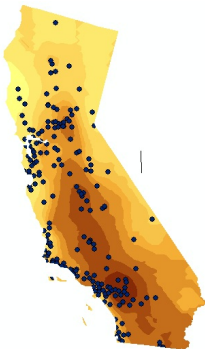
- ▶ maximum likelihood (model selection for fixed effects)
- ▶ REML (select covariance model, final fit)

Model and dimension	Functional form and parameter space
No-correlation $\mathbb{R}^d$	$\rho(r) = \begin{cases} 1 & \text{for } r = 0 \\ 0 & \text{for } r > 0 \end{cases}$
Triangular $\mathbb{R}$	$\rho(r; \alpha) = \begin{cases} 1 - \frac{r}{\alpha} & \text{for } 0 \leq r \leq \alpha \\ 0 & \text{for } r > \alpha \end{cases}$ $\alpha > 0$
Circular $\mathbb{R}^2$	$\rho(r; \alpha) = \begin{cases} \left(\frac{3}{2}\right) \left(\arccos\left(\frac{r}{\alpha}\right) - \frac{r}{\alpha} \sqrt{1 - \frac{r^2}{\alpha^2}} \right) & \text{for } 0 \leq r \leq \alpha \\ 0 & \text{for } r > \alpha \end{cases}$ $\alpha > 0$
Spherical $\mathbb{R}^3$	$\rho(r; \alpha) = \begin{cases} 1 - \frac{3r}{2\alpha} + \frac{r^3}{2\alpha^3} & \text{for } 0 \leq r \leq \alpha \\ 0 & \text{for } r > \alpha \end{cases}$ $\alpha > 0$



$$\begin{pmatrix} \mathbf{z}_{\text{observed}} \\ \mathbf{z}_{\text{unobserved}} \end{pmatrix} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \text{ var}(\boldsymbol{\epsilon}) = \boldsymbol{\Sigma}(\boldsymbol{\theta})$$

Prediction Map



Standard Error Map



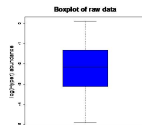
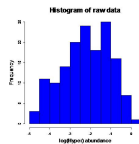
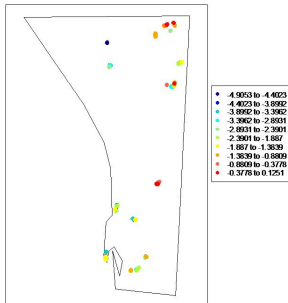


- ▶ Whiptail Lizard
- ▶ 148 locations in Southern California
- ▶ Measured the average number caught in traps over 80-90 trapping events in one year
- ▶ Data log-transformed, one outlier removed

Whiptail Lizard Data

$$\begin{pmatrix} \mathbf{Z}_{\text{observed}} \\ \mathbf{Z}_{\text{unobserved}} \end{pmatrix} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \text{ var}(\boldsymbol{\epsilon}) = \boldsymbol{\Sigma}(\boldsymbol{\theta})$$

- ▶ Ant Abundance
- ▶ Percent Sandy Soil
- ▶ Spherical Model
- ▶ Anisotropy



Fitted Model for Whiptail Lizard

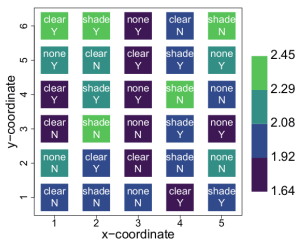
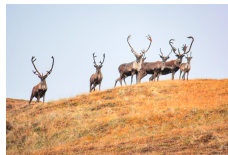
$$\begin{pmatrix} \mathbf{z}_{\text{observed}} \\ \mathbf{z}_{\text{unobserved}} \end{pmatrix} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \text{ var}(\boldsymbol{\epsilon}) = \boldsymbol{\Sigma}(\boldsymbol{\theta})$$

Effect	Est	Std Err	df	t-value	Pr(t):H ₀
Intercept	-3.994	0.5003	145	-7.984	< 0.00001
Ant Abund	0.306	0.1007	145	3.037	0.00283
Sandy Soil	1.080	0.2345	145	4.606	0.00001

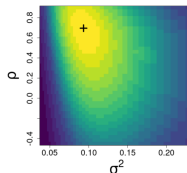
Parameter	Estimate
Partial Sill	0.723
Major Range	12.78
Nugget	0.524
Minor Range	0.012
Rotation	163.6

Spatial Design of Experiments

$$\begin{pmatrix} \mathbf{Z}_{\text{observed}} \\ \mathbf{Z}_{\text{unobserved}} \end{pmatrix} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \text{ var}(\boldsymbol{\epsilon}) = \boldsymbol{\Sigma}(\boldsymbol{\theta})$$



Source	df	Sum of Squares	Mean Square	F	P-value
Water	1	0.498	0.498	1.710	0.2033
Tarp	2	4.607	2.304	7.906	0.0023
Water × Tarp	2	0.756	0.378	1.297	0.2919
Error	24	6.993	0.291		
Total	29	12.855			



$$\begin{pmatrix} \mathbf{z}_{\text{observed}} \\ \mathbf{z}_{\text{unobserved}} \end{pmatrix} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \text{ var}(\boldsymbol{\epsilon}) = \boldsymbol{\Sigma}(\boldsymbol{\theta})$$

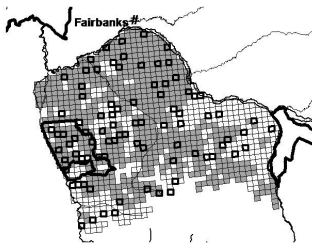
Total Area

SRS

- ▶ $\hat{\tau} = 11535$
- ▶ $\text{se}(\hat{\tau}) = 985$

Spatial Model

- ▶ $\hat{\tau} = 11327$
- ▶ $\text{se}(\hat{\tau}) = 978$



Small Area

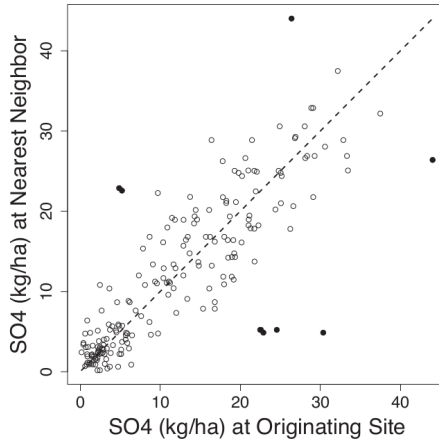
SRS(n=17)

- ▶ $\hat{\tau} = 1535$
- ▶ $\text{se}(\hat{\tau}) = 227$

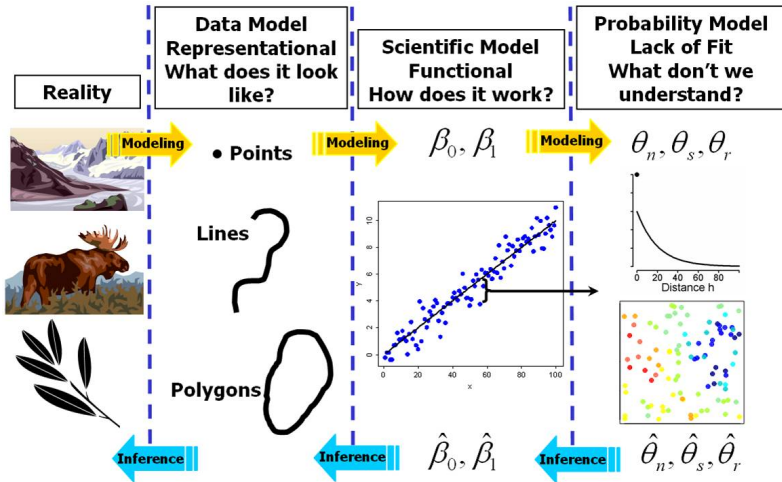
Spatial Model

- ▶ $\hat{\tau} = 1437$
- ▶ $\text{se}(\hat{\tau}) = 153$

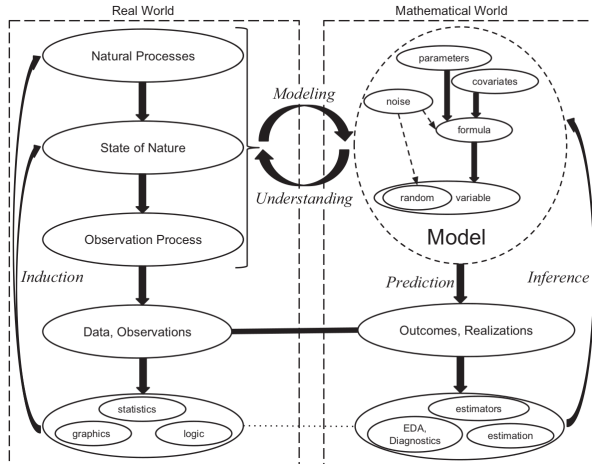
Model Diagnostics



Statistical Models and Inference



Model Selection



Ver Hoef, J.M. and Boveng, P.L. 2015. Iterating on a Single Model is a Viable Alternative to Multimodel Inference. *Journal of Wildlife Management* **79(5)**: 719-729

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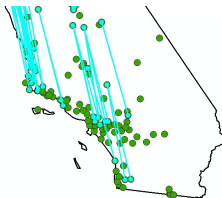
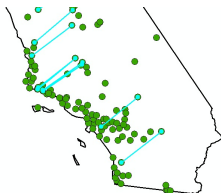
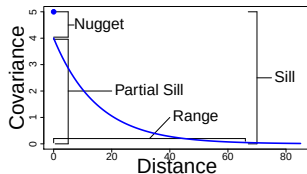
Model Selection Choices

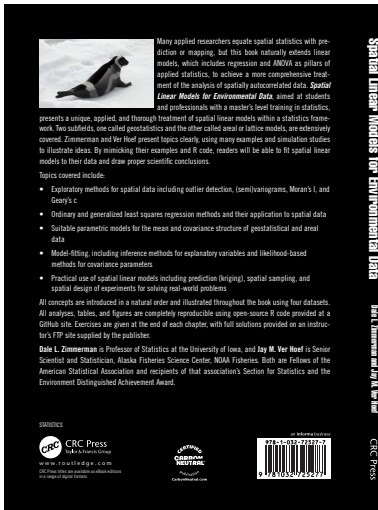
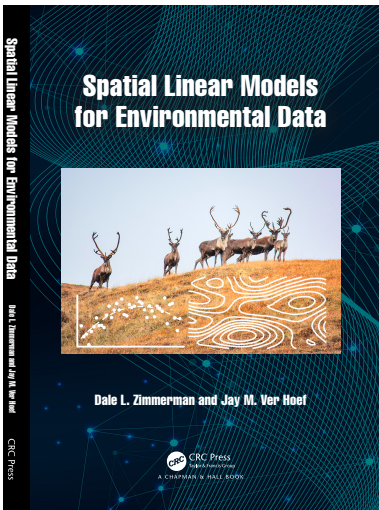
- ▶ Pick a single model
 - ▶ Design-based estimators, pre-registered
 - ▶ Very robust, correct α -level, not always efficient or powerful
- ▶ Iterate on a starting model
 - ▶ Forward or backward step-wise selection, by hand
 - ▶ Learn a lot, loose control of global α -level
- ▶ Choose a model from candidate set
 - ▶ Classical model selection
 - ▶ Inference on models, computationally expensive
- ▶ Multimodel inference,
 - ▶ model averaging
 - ▶ Makes sense for prediction, but maybe not for fixed effects

Metrics: P-values, AIC, BIC, cross-validation,
confidence/prediction interval coverage

Reduce the number of parameters in the probability model by using spatial relationships

$$\Sigma = \begin{pmatrix} \sigma_{1,1} & \sigma_{1,2} & \cdots & \sigma_{1,n} \\ \sigma_{2,1} & \sigma_{2,2} & \cdots & \sigma_{2,n} \\ \vdots & \vdots & \cdots & \vdots \\ \sigma_{n,1} & \sigma_{n,2} & \cdots & \sigma_{n,n} \end{pmatrix}$$





<https://usepa.github.io/spmodel/index.html>

spmodel 0.6.0 Reference Articles • Changelog

spmodel: Spatial Statistical Modeling and Prediction



Overview

spmodel is an R package used to fit, summarize, and predict for a variety of spatial statistical models applied to point-referenced and areal (lattice) data. Parameters are estimated using various methods, including likelihood-based optimization and weighted least squares based on variograms. Additional modeling features include anisotropy, non-spatial random effects, partition factors, big data approaches, and more. Model-fit statistics are used to summarize, visualize, and compare models. Predictions at unobserved locations are readily obtainable. Visit our website at <https://usepa.github.io/spmodel/>.

Installation

Install and load the most recent approved version from CRAN by running

```
# install the most recent approved version from CRAN
install.packages("spmodel")
# load the most recent approved version from CRAN
library(spmodel)
```

Install and load the most recent development version of spmodel from GitHub by running

```
# Installing from Github requires you first install the remotes package
install.packages("remotes")
```

Links

[View on CRAN](#)
[Browse source code](#)
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Citation

[Citing spmodel](#)

Developers

Michael Dumelle
 Author, maintainer

Matt Higham
 Author

Jay M. Ver Hoef
 Author

[More about authors...](#)

Dev status

Open issues: 0/0
 Open pull requests: 0/0
 Open discussions: 0/0
 Open milestones: 0/0

PLOS ONE

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RESEARCH ARTICLE

spmodel: Spatial statistical modeling and prediction in R

Michael Dumelle, Matt Higham, Jay M. Ver Hoef

Published: March 9, 2023 • <https://doi.org/10.1371/journal.pone.0282924>

Article	Authors	Metrics	CiteSpace	Media Coverage	Peer Review
1					

Abstract

spmodel is an R package used to fit, summarize, and predict for a variety of spatial statistical models applied to point-referenced or areal (lattice) data. Parameters are estimated using various methods, including likelihood-based optimization and weighted least squares based on variograms. Additional modeling features include anisotropy, non-spatial random effects, partition factors, big data approaches, and more. Model-fit statistics are used to summarize, visualize, and compare models. Predictions at unobserved locations are readily obtainable.

The spatial linear model

Model fitting

Prediction

Advanced features

Discussion

Acknowledgments

References

Reader Comments

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Introduction to Spatial Modeling with spmodel

https://highamm.github.io/spmodel_intro/